Chalumuri Shiva Rohit

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.95
SRI CHAITANYA TECHNO SCHOOL VISAKHAPATNAM AP	Class XII (CBSE)	2024	93.60%
SRI CHAITANYA EM SCHOOL RAMNAGAR,VSP VISAKHAPATNAM DISTRICT	Class X (SSC)	2022	96.83%

PROJECTS

Arbitrary Precision Arithmetic Library :

- Built a Java library for arbitrary precision integer and floating point arithmetic using string based arithmetic instead of built in numeric types removing their overflow limitation

- Implemented manual digit wise algorithms for addition, subtraction, multiplication, and division with an object oriented design and reusable helper methods.

- GitHub link : https://github.com/rohitshiva2310/arbitrary-precision-arithmetic

C code conversion for integer operation to use FP operations to allow speedup : (Group project : 50 % contribution)

- Used gem5 on intel systems to simulate the execution of various C programs in both integer and mixed version having both integer and floating point operations to see if any decrement in execution time is possible

- The programs include factorial evaluation , matrix multiplication , polynomial evaluation and were done for various values and variations of workload among integer and floating point units.

- GitHub link : https://github.com/rohitshiva2310/integer-fp-optimisation

Disk Scheduling and RAID-backed swap in xv6 OS :

- Extended the xv6 operating system with page fault handling , clock algorithm for page replacement, and swap space management in the kernel

- Designed a RAID backed swap subsystem supporting RAID 0/ RAID 1/ RAID 5 with FCFS and SSTF disk scheduling, including fault recovery and RAID-5 reconstruction

- Built kernel level virtual memory statistics and system calls to monitor page faults, swap operations , disk operations, and latency

-GitHub link : https://github.com/rohitshiva2310/xv6-raid-disk-scheduler

Web based Karnaugh Map Solver and Verilog Generator (Group project : 25% contribution)

- Developed a website that minimizes Boolean expressions from user defined truth tables, visualizes a karnaugh map and supports 2-4 variable logic expression simplification

- Implemented automatic verilog module and testbench generation from the minimized logic expression

- GitHub link : https://github.com/rohitshiva2310/kmap-solver

SKILLS

Programming and scripting languages : C, C++ , Python * , SQL* , HTML , CSS , JavaScript * , Assembly (RISC-V)

Libraries and frameworks : STL

Tools : Git , GitHub , LATEX

- (* is for elementary proficiency)

RELEVANT COURSES

Computer Science Fundamentals : Introduction to Programming , Data Structures , Algorithms , Operating systems , Database management systems , Computer Architecture and organisation , Theory of computation , Software development fundamentals.

Mathematics : Linear algebra , Probability and random variables , statistics, elementary calculus

CODING / CONFERENCE

Leetcode : Solved around 300 plus problems of different topics

- profile link : https://leetcode.com/u/rohit_shiva23/

Codeforces : Solved over 200 problems across various topics .

- profile link : https://codeforces.com/profile/rohit_shiva

ACHIEVEMENTS AND AWARDS

- Participated in various olympiads like KAT (Knowledge assessment test) and NSTSE in school level and at intermediate college level and received some awards in them after going to last rounds